GRAVITYSORT · DESIGN DECISION RECORD

GravitySort recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GVS-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/gravitysort		
Source	docs/decisions/0002-recommendations-accepted.md @ 6c7e5bc		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 8b, 11b, 12 and 13 decided by Amish as recommended (GVS-DEC-001)
0.2	2026-09-26	Amish Chadha	Budget approved by Amish ($455)
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of the recommendations in GVS-DDR-001 and docs/REVIEW.md, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted. Every item in GVS-DDR-001 and `docs/REVIEW.md` that carried a recommendation is decided by Amish, 2026-09-25: go with recommendation. Items 8b, 11b, 12 and 13, which had no recommendation, were decided by Amish on 2026-10-02 as later recommended: "i approve your recommendations for all 555 open decisions." (GVS-DEC-001).

Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." GVS-DDR-001 (v0.1) had adopted most TRL 2 recommendations for TRL 3 work, open for his review, and listed the budget and six TRL 3 items as awaiting him. This record turns every item that had a recommendation into a decision, applies it at TRL 3 level inside this repo, and lists what is still open. Where a recommendation offered several options, the recommended option is the decision. TRL 4 work (building, testing, purchasing) stays on hold by Amish's instruction, and the project stays at TRL 3.

Decision

Table 1. Items decided by Amish, 2026-09-25: go with recommendation.

# (GVS-DDR-001)	Decision	What changed in the repo
1	Budget raised from $350 to **$450**	`project.yaml` `budget_usd` 350 to 450; R9 target $450 (GVS-REQ-001 v0.4); README budget line; GVS-CAL-001 v0.2 section 12. The BOM is $455, so R9 is still not met, by $5
2	Two stages: centrifugal bowl, then shaking table	Status wording only (GVS-PRC-001 v0.4, GVS-DDR-001 v0.2)
3	Fluidized bowl with a rotary union; non-fluidized bowl kept as a documented low-cost variant	Status wording only
4	Cast polyurethane liner in a 3D-printed mold on a GFRP shell	Status wording only
5	Pedal crank baseline, MotionCore 250 W option, one drive time-shared between bowl and table	Status wording only
6	Battery for the motor option left to the user; SwapCell not named	Status wording only
7	Direct smelting with borax as the downstream step, outside this hardware	Status wording in GVS-PRB-001 v0.4
8a	Partner screening criterion: a country with a mercury ban in force (Colombia) or a Minamata action plan with a planetGOLD programme	Status wording in GVS-PRB-001 v0.4
9	R12 restated (option a): with the pedals, a burst safety factor of 10 or more at the highest reachable speed (1,200 rpm at a 100 rpm cadence) plus the speed display replaces the gearing cap; the motor stays capped at 900 rpm	GVS-REQ-001 v0.4 R12; GVS-CAL-001 v0.2 sections 10 and 14 and `sizing.py`. R12 moves from not met as written to **met on paper** (lowest factor 17)
10	Lighter frame (option b): 25 x 25 x 1.5 mm tube for the base frame, pedal outrigger, table stretchers and table head post, then re-estimate	`cad/src/model.py` (`tube` 30 to 25, new `tube_wall` 1.5; cross members and stand stretchers follow the tube size); STEP and STL re-exported; GVS-DWG-001 Rev P1 to P2; BOM items 1 ($40 to $32) and 15 ($30 to $28); GVS-CAL-001 v0.2 adds a frame member check (61 MPa, safety factor 3.9, 0.73 mm deflection). Mass 88.3 kg to **77.3 kg**, heaviest load 25.8 to 18.1 kg: R11 **met on paper**
11a	Fluidization supply: 3/4 in hose and about 89 holes of 1.0 mm graded toward the lower rings	Already in the model notes and BOM; status wording in GVS-PRC-001 v0.4 and GVS-CAL-001 v0.2 section 6
14	Engineering proposals: disc brake with parking latch and lid interlock pin (BOM item 18); bicycle computer speed display (item 19); 1.2:1 motor chain step-up; MotionCore lid switch on a brake input	Already in the model, BOM and calculation; status wording in GVS-PRC-001 v0.4. The MotionCore assumptions are listed as cross-repo actions in `docs/REVIEW.md`

No pitch or problem rewording was recommended, so the `project.yaml` pitch and problem are unchanged.

Items left open on 2026-09-25

Table 2. Items with no recommendation on 2026-09-25, and their status since.

# (GVS-DDR-001)	Item	Why it was open	Status
1 (part)	Cost-down options to close the $5 gap: quarter-turn belt instead of the bevel gearbox (about $25), non-fluidized variant (about $45), table later (about $70 deferred)	Listed as options with no recommendation	Carried as savings worth trying in the Value engineering section of GVS-DEC-001, not as an open decision
8b	The first co-design partner itself	No partner was recommended; only the screening criterion	Decided by Amish, 2026-10-02, as recommended: Colombia, with a miners' cooperative introduced through the Alliance for Responsible Mining in Medellin as the first candidate (GVS-DEC-001)
11b	Head for the fluidization supply: a 1.7 m post or a small pump, for 10 kPa net at every ring	No preference was stated between the two	Decided by Amish, 2026-10-02, as recommended: a 1.7 m post, braced (GVS-DEC-001)
12	Water pumping from the settling pond at pedal-only sites (about 1.19 m3/h, 5.5 W hydraulic)	No recommendation yet	Decided by Amish, 2026-10-02, as recommended: gravity supply from upstream as the site rule; otherwise a treadle or hand pump, or a 12 V pump with the MotionCore battery (GVS-DEC-001)
13	R2 and R5 margins: raise the design feed to about 210 kg/h, or restate R2 as 200 kg/h of feed time	Two options, no recommendation	Decided by Amish, 2026-10-02, as recommended: R2 restated as 200 kg/h of feed time; design feed kept at 200 kg/h (GVS-DEC-001)

On hold (TRL 4)

Nothing decided here needs TRL 4 work to be recorded. The checks the decisions point at, such as bearing alignment on the first 25 mm frame, spin-up with the tub and lid as containment, and brake stop time, are TRL 4 work and stay on hold by Amish's instruction.

Consequences

- GVS-PRB-001, GVS-PRC-001 and GVS-REQ-001 move to v0.4, GVS-CAL-001 to v0.2 and GVS-DDR-001 to v0.2, each with a revision entry for this record.
- Requirements then stood at one not met (R9, $455 against $450), three at risk (R2, R5, R6), two not verifiable before testing (R4, R14) and eight met on paper (R1, R3, R7, R8, R10, R11, R12, R13).
- The pedal drive is still not speed-capped; the restated R12 makes the structural margin and the speed display carry that risk, and the tub and lid are not yet shown to contain a liner fragment.
- `trl` and `trl_target` stay at 3.

Budget approved, 2026-09-26

On 2026-09-26 Amish wrote, in chat: "i approve all the budget items."

- Budget set to $455 to cover the priced BOM: decided by Amish, 2026-09-26. This closes the $5 gap left by item 1. The BOM is $455 (MotionCore and battery excluded), so R9 moves from not met to met on paper, with no margin. The cost-down options in GVS-DDR-001 item 1 stay on record but are no longer needed.
- Requirements now stand at none not met, three at risk (R2, R5, R6), two not verifiable before testing (R4, R14) and nine met on paper (R1, R3, R7, R8, R9, R10, R11, R12, R13).
- Files changed: `project.yaml` (`budget_usd` 450 to 455); GVS-REQ-001 v0.5; GVS-CAL-001 v0.3, `docs/04-calcs/sizing.py` (hard-coded budget 450 to 455) and `results.csv` ; GVS-PRB-001 v0.5 and GVS-PRC-001 v0.5 (budget figure); `README.md` ; `bom/bom-notes.md` ; `docs/REVIEW.md` .